The subscript p is just a reminder that $\psi_p(x)$ is the eigenvector of **P** with the specific eigenvalue p. It is a function of x, but it is labeled by an eigenvalue of **P**.

The constant A multiplying the exponential is not determined by the eigenvector equation. That's nothing new; the eigenvalue equation never tells us the overall normalization of the wave function. As a rule, we fix the constant by requiring the wave function to be normalized to unit probability. An example that goes all the way back to Section 2.3 is the eigenvector of the x component of spin:

$$|r\rangle = \frac{1}{\sqrt{2}}|u\rangle + \frac{1}{\sqrt{2}}|d\rangle.$$

The factor $1/\sqrt{2}$ is there to make sure the total probability is 1.

Normalizing the eigenvectors of **P** is a more subtle operation, but the result is simple. The factor A is only slightly more complicated than in the spin case. To save time, I will tell you the answer and leave it for you to prove later. The correct factor is $A = 1/\sqrt{2\pi}$ Thus,

$$\psi_p(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{ipx}{\hbar}}. \qquad (8.17)$$

A point of some interest follows from Eqs. 8.13 and 8.17. The inner product of a position eigenvector $|x\rangle$ and a momentum eigenvector $|p\rangle$ has a very simple and symmetric form: